



Convex Optimization

Project 3

spring 1403
Due date: 21th of Ordibehesht



Flight simulator. A flight simulator moves in a bounded region, attempting to recreate the pilot's feel of the motion of an airplane. In this problem we consider a simple 1D version of this problem. (Real flight simulators involve 6 degrees of freedom.) Let $p^{\text{ref}} \in \mathbf{R}^T$ denote the (1D) position of the airplane at time periods $t = 1, \dots, T$, where time periods are spaced $h > 0$ seconds apart. The velocity is given by $v^{\text{ref}} \in \mathbf{R}^{T-1}$,

$$v_t^{\text{ref}} = \frac{p_{t+1}^{\text{ref}} - p_t^{\text{ref}}}{h}, \quad t = 1, \dots, T-1,$$

the acceleration $a^{\text{ref}} \in \mathbf{R}^{T-2}$ is

$$a_t^{\text{ref}} = \frac{v_{t+1}^{\text{ref}} - v_t^{\text{ref}}}{h}, \quad t = 1, \dots, T-2,$$

and the jerk (third derivative) $j^{\text{ref}} \in \mathbf{R}^{T-3}$ is

$$j_t^{\text{ref}} = \frac{a_{t+1}^{\text{ref}} - a_t^{\text{ref}}}{h}, \quad t = 1, \dots, T-3.$$

You will choose the position of the simulator, $p \in \mathbf{R}^T$, which must satisfy $\|p\|_{\infty} \leq P^{\text{max}}$, where P^{max} is given. We let v , a , and j denote the velocity, acceleration, and jerk of the simulator, defined in the same way as above for the reference. A person's sense of motion is sensitive to jerk, which motivates the simple least squares penalty

$$\sum_{t=1}^{T-3} (j_t - j_t^{\text{ref}})^2.$$

The data in the problem are p^{ref} , h , and P^{max} .

(a) Explain how to solve this problem using convex optimization, i.e., find the simulator position trajectory $p \in \mathbf{R}^T$ that minimizes the penalty described above subject to the constraints described above.

(b) Solve the problem with data given in `flight_simulator_data.py`. Plot the optimal simulator position p^* , and compare it to the reference position p^{ref} . Plot the jerk of the simulator and the reference on the same plot.

Good Luck!